```matlab
% Drone Thrust-to-Weight Analysis

% Constants
ThrustPerMotor = 1; % In kg
Gravity = 9.81; % In m/s^2
BaseDroneMass = 1; % In kg
NumberOfMotors = 4;
Volume = 0.000029; % In m^3

T = readtable('materials.csv');
materials = table2struct(T);

% Arrays to store results
MaxPayload = zeros(length(materials),1);
ThrustToWeightRatio = zeros(length(materials),1);
TotalMass = zeros(length(materials),1);

for i = 1:length(materials)

    Density = materials(i).rho_kg_m3;

    % Arm Mass
    ArmMass = Volume * Density; % In kg

    % Payload
    Payload = 0.5; % Minimum Required Payload

    % Total Drone Mass
    TotalMass(i) = BaseDroneMass + (NumberOfMotors*ArmMass) + Payload;

    % Total Thrust
    TotalThrust = (NumberOfMotors*ThrustPerMotor);

    % Thrust-To-Weight
    ThrustToWeightRatio(i) = TotalThrust/TotalMass(i);

    % Maximum Allowable Mass
    MaxTotalMass = TotalThrust/2; % Incorporates the 2:1 ratio

    % Maximum Payload
    MaxPayload(i) = MaxTotalMass - BaseDroneMass - (NumberOfMotors*ArmMass);

    % Show Results
    fprintf('\n==============================\n');
    fprintf('Material %d\n',i);
    disp(materials(i))

    fprintf("Total Drone Mass: %.3f kg\n", TotalMass(i));
    fprintf("Maximum Payload: %.3f kg\n", MaxPayload(i));
    fprintf("Thrust-to-Weight Ratio: %.2f\n", ThrustToWeightRatio(i));
```

```matlab
    % Check Requirements
    if ThrustToWeightRatio(i) >= 2 && MaxPayload(i) >= 0.5
        disp("Design meets the requirements")
    else
        disp("Design does not meet the requirements")
    end

end
```

```
=============================
Material 1
                name: 'Carbon Fiber Composite (CFRP)'
           rho_kg_m3: 1600
                E_Pa: 7.0000e+10
                  nu: 0.3000
    yieldStrength_Pa: 600000000
      cost_USD_per_m: 30
Total Drone Mass: 1.686 kg
Maximum Payload: 0.814 kg
Thrust-to-Weight Ratio: 2.37
Design meets the requirements
=============================
Material 2
                name: 'Aluminum Alloy'
           rho_kg_m3: 2700
                E_Pa: 6.9000e+10
                  nu: 0.3300
    yieldStrength_Pa: 275000000
      cost_USD_per_m: 7.5000
Total Drone Mass: 1.813 kg
Maximum Payload: 0.687 kg
Thrust-to-Weight Ratio: 2.21
Design meets the requirements
=============================
Material 3
                name: 'Fiberglass Composite (GFRP)'
           rho_kg_m3: 1900
                E_Pa: 2.5000e+10
                  nu: 0.2800
    yieldStrength_Pa: 300000000
      cost_USD_per_m: 15
Total Drone Mass: 1.720 kg
Maximum Payload: 0.780 kg
Thrust-to-Weight Ratio: 2.33
Design meets the requirements
=============================
Material 4
                name: 'PLA Plastic'
           rho_kg_m3: 1250
                E_Pa: 3.5000e+09
                  nu: 0.3600
    yieldStrength_Pa: 60000000
      cost_USD_per_m: 3.5000
Total Drone Mass: 1.645 kg
Maximum Payload: 0.855 kg
Thrust-to-Weight Ratio: 2.43
Design meets the requirements
=============================
Material 5
                name: 'ABS Plastic'
           rho_kg_m3: 1050
```

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              E_Pa: 2.0000e+09
                nu: 0.3500
  yieldStrength_Pa: 40000000
    cost_USD_per_m: 3.5000
Total Drone Mass: 1.622 kg
Maximum Payload: 0.878 kg
Thrust-to-Weight Ratio: 2.47
Design meets the requirements
=============================
Material 6
              name: 'Wood (Birch)'
         rho_kg_m3: 600
              E_Pa: 1.0000e+10
                nu: 0.3500
  yieldStrength_Pa: 80000000
    cost_USD_per_m: 2
Total Drone Mass: 1.570 kg
Maximum Payload: 0.930 kg
Thrust-to-Weight Ratio: 2.55
Design meets the requirements
```

```
% Summary Table
ResultsTable = table(...
    (1:length(materials))',...
    TotalMass,...
    MaxPayload,...
    ThrustToWeightRatio,...
    'VariableNames',
{'Material','TotalMass_kg','MaxPayload_kg','ThrustToWeightRatio'});

disp(ResultsTable)
```

Material	TotalMass_kg	MaxPayload_kg	ThrustToWeightRatio
1	1.6856	0.8144	2.373
2	1.8132	0.6868	2.206
3	1.7204	0.7796	2.325
4	1.645	0.855	2.4316
5	1.6218	0.8782	2.4664
6	1.5696	0.9304	2.5484